

**ERC Starting Grant 2026
Part B1¹**

**Multisensory Unsupervised Statistical Inference in Cortical
Circuits: Learning Associations Across Senses
MUSIC**

Cover Page:

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- Name of the PI's Host Institution for the project: CHUV
- Proposal duration in months: 60

Proposal Summary:

MUSIC (Multisensory Unsupervised Statistical Inference in Cortical circuits) investigates how early postnatal audiovisual experience shapes cortical circuits and perception. Perception is not a passive reflection of sensory input but is strongly influenced by expectations distilled from prior experience. In natural environments, specific sensory signals often co-occur across modalities, yet it remains poorly understood how sustained exposure to such cross-modal regularities alters cortical processing and enables the learning of multisensory associations. MUSIC will test the hypothesis that early exposure to experimentally imposed audiovisual correspondences can induce predictable effects on perception, reshape neural codes, and leave identifiable synaptic traces. To this end, juvenile mice will be reared in environments where natural scenes are systematically paired with algorithmically generated sounds, creating arbitrary but learnable audiovisual mappings. Behavioral assays will assess how, following this experience, the same sounds may bias visual decisions towards the associated visual features, while miniaturized head-mounted two-photon imaging will link these behavioral effects to spontaneous and evoked activity changes in visual and associative cortices. Dendritic spine imaging will explore the synaptic substrate of these associations, and computational models will provide quantitative predictions. By bridging perception, neurophysiology, and theory, MUSIC will uncover how audiovisual learning builds cross-modal perceptual priors, advancing our understanding of unsupervised statistical learning in the brain and paving the way for clinical interventions that rely on artificial multisensory learning, such as neuroprosthetics and sensory substitution.

¹ Instructions for completing Part B1 can be found in the 'Information for Applicants to the Starting and Consolidator Grant 2026 Calls'.

Background and rationale

Statistical and unsupervised learning in sensory systems

A central theme in sensory neuroscience is that the identity and distribution of stimuli we encounter in our environment (i.e. their *statistics*) play a fundamental role in shaping how sensory inputs are encoded in neural activity^{1,2}. The sensory environment we are exposed to, particularly during critical periods of early development³, profoundly influences how we perceive and interpret the world. Classic examples illustrate this principle: we become more proficient at distinguishing the phonemes of the languages heard in early life⁴, we recognize faces more accurately from ethnicities commonly encountered in childhood^{5,6}, and we are differentially sensitive to geometric illusions depending on the visual environments in which we grow up⁷. Such findings underscore the brain's remarkable ability to internalize the statistical structure of experience, molding perceptual systems in ways that persist across the lifespan.

This process is thought to arise largely from passive exposure to sensory experience, without requiring explicit feedback, an “unsupervised” form of learning that builds on learning rules capturing correlations between sensory features². This idea stems from early neurobiological theories of learning and plasticity, such as Hebb's rule (“neurons that fire together wire together”). In recent years, unsupervised learning has gained renewed attention in both neuroscience and machine learning, offering a promising route not only to more powerful artificial intelligence systems but also to a deeper understanding of the organizing principles of brain function^{2,8}.

In parallel, computational neuroscience has developed theories of statistical inference, describing how the brain combines expectations distilled from prior experience with incoming sensory evidence to achieve robust perception^{9,10}. Rooted in Bayesian statistics, these frameworks have been highly successful in explaining both human and animal behavior^{9,10}. They have also inspired influential hypotheses, for example, that the activity of cortical circuits provides a direct window onto the statistical structure of sensory stimuli internalized through experience¹⁰.

Neurobiological evidence strongly supports the relevance of these ideas. Statistical and unsupervised learning offer not only normative frameworks to explain many characteristics of adult sensory systems^{2,11}, but also a predictive model of how exposure to altered environments induces change during both development and adulthood². For example, in the early postnatal period³, the spatial and temporal statistics of sensory input shape the selectivity of single neurons¹², the stability of their responses to visual stimuli¹³, and even the structure of spontaneous activity in primary visual cortex (V1)¹⁴. At the other end of the spectrum, rapid adaptive changes can also occur in adulthood: in the high order visual cortex of monkeys, less than one hour of unsupervised exposure to altered visual statistics (e.g., object identity swaps during saccades) was sufficient to reshape neuronal selectivity in a location-specific manner¹⁵. Similarly, in the human temporal lobe, the rapid formation of contextual associations between concepts has been shown to alter the firing patterns of single neurons¹⁶.

Learning cross-modal associations and multisensory priors

Beyond unimodal sensory learning, humans are also exceptionally adept at extracting associations across modalities^{17–19}. Hearing a dog bark instantly evokes its visual image; the clang of a gong conjures the sight of a bronze plate. These intuitive, automatic correspondences across senses (i.e. cross-modal associations) emerge largely through experience and vary across individuals and cultures. Because many such correspondences vary over relatively short timescales and across environments, far faster than evolutionary change, they are unlikely to be genetically hard-wired and therefore must be learned from experience²⁰. In the language of statistical inference, these associations can be seen as multisensory *priors* representing *expectations* on the content of one sensory modality given another one. The brain's capacity for this type of learning represents a powerful mechanism by which the cognitive systems build internal models of the world, exploiting correlations between disparate sensory inputs to generate expectations that guide robust perception¹⁸. Studying the learning of multisensory associations is important not only in its own right, with clear implications for neurological disorders and clinical interventions, but also as an invaluable window into the general mechanisms of sensory learning and inference described above. Unlike unimodal settings, multisensory paradigms and in particular audiovisual ones, allow for experimentally convenient manipulation of co-occurrence statistics of stimuli between modalities, while at the same time enabling clearer tracking of their neural processing within each modality. This is made possible by the existence of relatively segregated sensory pathways in the brain, which can be selectively monitored and modelled²¹. Moreover, the delayed developmental trajectory of multisensory processing³ makes it possible to study the underlying learning mechanisms systematically in both juvenile and adult subjects. In this way, multisensory statistical learning provides a uniquely powerful framework to dissect how experience shapes cortical circuits and builds the priors that underlie perception.

Empirical evidence on audiovisual learning in cortical circuits

A rich body of work on the mechanisms of multisensory integration has shown that concurrent auditory and visual stimulation can, in some contexts, sharpen tuning^{22,23} in visual cortex by recruiting both excitation and inhibition²⁴, in a manner that depends on the degree of feature matching across modalities (i.e., *congruency*)²⁵. This is consistent with the idea that cortical circuits implement a form of statistical inference based on the natural co-occurrence of sounds and visuals. However, such findings do not directly probe the brain's ability to learn new associations from experience. So far, only a limited number of studies have directly addressed specifically the circuit basis of audiovisual learning. It has been shown, for example, that reward-based training can reshape auditory and visual responses in mouse cortex²⁶, that short-term passive exposure to simple audiovisual pairings can alter audiovisual tuning in V1²⁷ and, that in superior colliculus, spatiotemporal audiovisual integration rules are calibrated by early experience^{20,28}. Yet we still lack an integrated account of how prolonged exposure to specific audiovisual statistics, particularly during development, modifies perception and sensory coding in the mammalian cortex. MUSIC will bridge this gap.

The overarching hypothesis

The central hypothesis of MUSIC is that extended exposure to experimentally imposed audiovisual correspondences during the early postnatal period will induce changes in the sensory cortex of mice that reflect the build-up of an internal model of the altered cross-modal statistics of the environment (Figure 1). This internal model should act during perceptual inference as a multisensory prior and manifest across multiple levels:

- 1) **Perception**: as cross-modal biases in perceptual decision-making (e.g., auditory contexts shifting psychometric curves in the direction of the paired visual features).
- 2) **Neural activity**: as changes in both evoked and spontaneous cortical responses, including sound-evoked “visual-like” activity in darkness, as predicted by sampling-based theory of statistical inference in cortex.
- 3) **Synaptic architecture**: as selective and localized plasticity in dendritic spines of visual cortical neurons, physically embedding the learned audiovisual associations in the synaptic micro-architecture of the cortex.

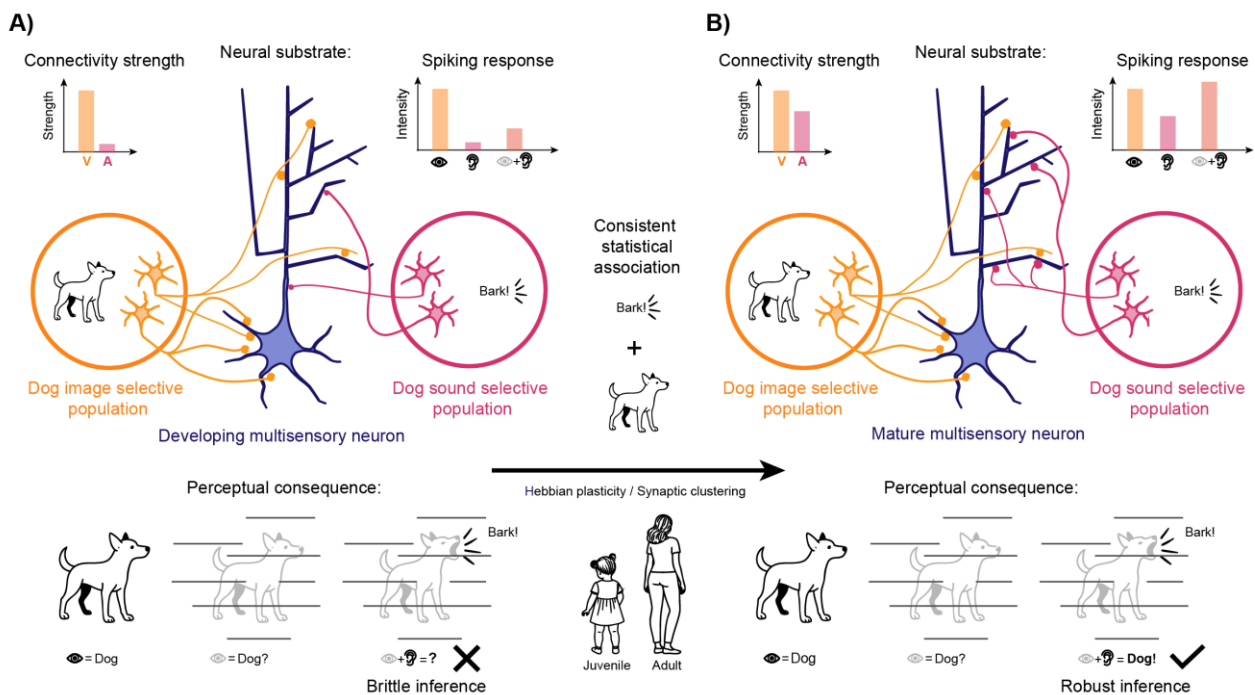


Figure 1 | Conceptual model of audiovisual statistical learning

Conceptual model of audiovisual statistical learning. A) Before exposure: developing multisensory neurons have very asymmetrical connectivity across modalities, leading to weak audiovisual integration and unreliable perceptual outcomes. B) After exposure: Hebbian plasticity strengthens congruent connections on a developmental timescale leading to mature, strongly integrative, multisensory neurons able to support reliable multisensory inference and ensure robust perception (e.g. of a dog in a foggy day).

Together, these experiments will test whether unsupervised developmental exposure to multisensory statistics is sufficient to forge lasting cross-modal priors, clarifying the principles by which the brain constructs multisensory inferences from experience.

Objectives and work plan

WP1 - Impact of unsupervised audiovisual exposure on perception and behavior

Design. Juvenile mice will grow up in immersive environments where what they see and hear are systematically linked in an arbitrary and experimentally controlled way. For several hours each day during post-weaning development, natural scenes displayed on surrounding monitors will be paired with corresponding soundscapes delivered from speakers. Each image will be converted into a sound whose pitch reflects its dominant orientation content, for instance, scenes rich in horizontal features (like the dunes in Figure 2A, right) will be accompanied by lower-pitched sounds, whereas vertical-rich scenes (like the forest in Figure 2A, left) will co-occur with higher-pitched sounds. In this way, specific visual features are consistently experienced together with specific sounds, providing ample opportunity for the animals to learn these cross-modal correspondences (Figure 2A). In adulthood, these animals will be tested to determine whether their early sensory environment has left a measurable imprint on perception. In a freely moving two-alternative forced-choice task, mice will judge the orientation of visual stimuli while occasionally hearing sounds that either match or mismatch the visual stimulus presented during their rearing. If the audiovisual association has been internalised as a perceptual cross-modal prior, then, particularly on trials where the visual input is ambiguous (e.g. low discriminability), the sound should bias perceptual judgements toward the associated orientation, shifting the psychometric curve²⁹ (Figure 2B). Control cohorts will be reared with identical visual and auditory statistics but with random pairings, ensuring that any perceptual bias reflects learned audiovisual structure rather than generic exposure. As an independent assay, second-order cross-modal fear conditioning³⁰ will test whether rearing-induced associations are strong enough for sounds alone to reinstate freezing in darkness after conditioning with the corresponding visual stimulus.

Predictions and impact. We expect sounds to bias visual choices toward their associated orientation, producing measurable shifts in psychometric curves and in second-order conditioning. Such effects would demonstrate that early-life audiovisual statistics are internalised as cross-modal perceptual priors influencing adult decisions. A Bayesian observer model will test whether cue integration is statistically optimal^{9,31}, while artificial neural networks trained on the same audiovisual “diet” will generate mechanistic predictions (Figure 2E). Pilot computational modelling already predicts strong biases, supporting feasibility. Observing the predicted effect would constitute a landmark causal demonstration that early-life multisensory experience plays an instructive role in shaping adult perceptual inference in mammals.

WP2 - Impact of audiovisual exposure on evoked and spontaneous cortical activity

Design. To uncover how cross-modal perceptual priors from WP1 are implemented in cortical circuits, we will image neuronal population activity in the visual cortex of freely moving mice during both perceptual decision-making and passive playback of auditory stimuli in darkness (Figure 2C). Using miniaturised head-mounted two-photon microscopes³² (Mini2P), we will record calcium signals^{33,34} (GCaMP6f) from hundreds of neurons in V1 and in higher-order associative areas implicated in audiovisual integration (such as the anterolateral area, AL). Two key predictions will be tested:

WP2.1 - Evidence integration. During low-discriminability trials in the task, rearing-consistent sounds should bias visual cortical representations toward the associated orientation. Population decoders trained on high-contrast trials will be applied to low-discriminability conditions to assess whether neural activity shifts toward the representation of the sound-associated stimulus. These neurometric shifts are expected to parallel the psychometric biases observed in WP1²⁹ (Figure 2B).

WP2.2 - Prior sampling. In complete darkness, when no visual input is available, playback of rearing sounds should evoke population activity patterns resembling those normally elicited by their paired visual stimuli, indicating that “spontaneous” visual cortical dynamics can express samples from learned priors¹⁰, conditioned by the current auditory input.

As in WP1, control cohorts will be reared with identical visual and auditory statistics but with random pairings, ensuring that any observed neural effects specifically reflect the altered cross-modal structure of the environment.

Predictions and impact. We expect audiovisual rearing to leave measurable traces in cortical population codes. During task performance, rearing-consistent sounds should bias neural representations in visual cortex toward the associated orientation, providing the neural substrate for the behavioral effects observed in WP1. In darkness, the same sounds should evoke visual-like activity patterns interpretable as cross-modally induced imagery of the previously paired stimuli³⁵, or as samples from the learned multisensory prior, as predicted by sampling-based theories of perceptual inference^{10,14}. Confirming or refuting this prediction would have broad implications for understanding how the cortex implements probabilistic inference. As in WP1, artificial neural networks trained on the same audiovisual “diet” will generate model predictions of representational bias, allowing direct comparison between brains and machines⁸ (Figure 2E). Pilot Mini2P

recordings have already demonstrated multisensory selectivity in associative cortices during object free exploration, confirming the feasibility of the approach (Figure 2C). Together, these experiments will establish the cortical population substrate of experience-dependent cross-modal priors, illuminating the neural mechanisms of multisensory perceptual inference.

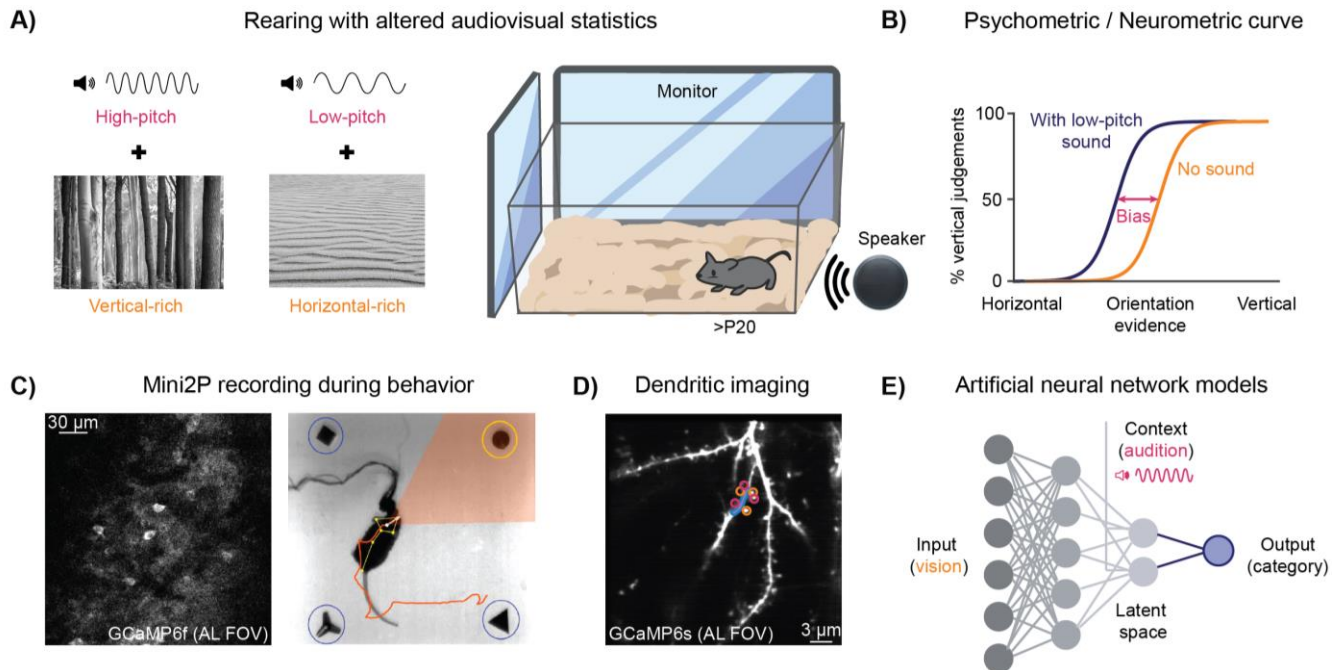


Figure 2 | MUSIC project overview

A) Experimental rearing paradigm with altered audiovisual statistics: animals will be raised in a sensory environment where images rich in specific visual orientation content are consistently paired with specific sound frequencies. **B) Behavioral readout:** animals exposed to the altered statistics will be trained in two-alternative forced-choice (2AFC) visual discrimination tasks to test whether auditory context biases visual orientation judgements, shifting psychometric curves. **C) Neural readout:** population-level recordings during freely moving behavior will be performed with the Mini2P to investigate the neural substrates of the learned perceptual biases (field of view from freely behaving mouse, pilot Mini2P recordings). **D) Synaptic substrates:** dendritic two-photon imaging will be used to test for the presence of clustering among congruently tuned synaptic inputs, potentially forming the physical substrate of the learned multisensory associations (dendrite from pilot two-photon recordings). **E) Artificial neural network models:** networks trained with unsupervised learning algorithms will quantitatively link audiovisual statistics to perceptual decisions, generating predictions directly comparable to mouse behavior and neural activity.

WP3 - Synaptic substrates of audiovisual statistical learning

Design. To identify where and how multisensory associations are stored at the level of synaptic connections, we will image dendrites, the elementary sites where different synaptic inputs converge onto a neuron, after audiovisual rearing. Using two-photon microscopy in head-fixed mice (for optimal mechanical stability) expressing a calcium indicator (GCaMP6s³³) and a structural marker (e.g. mRuby)³⁶, we will record activity from individual dendritic spines in visual cortex while presenting visual and auditory stimuli (Figure 2D). By measuring local calcium transients, we will reconstruct the tuning of each spine to specific visual orientations and sound frequencies along short stretches of dendrite.

This approach will test a simple but fundamental hypothesis: do synaptic inputs carrying congruent (i.e. statistically associated during early postnatal experience) audiovisual information tend to cluster together along the same dendritic branches? If so, such clustering would provide a physical substrate for the associative plasticity expected to arise from prolonged exposure to structured audiovisual statistics, potentially enabling nonlinear amplification³⁷ of congruent inputs and turning the neuron into an integrative multisensory unit (Figure 1). Control cohorts reared with identical unimodal statistics but random pairings will confirm that any observed clustering is specific to the specific audiovisual co-occurrences experienced in the early postnatal period. Preliminary two-photon imaging has already demonstrated the feasibility of reconstructing individual spine tuning to sensory features from different modalities in mouse visual cortex, providing a solid foundation for this experiment.

Predictions and impact. We expect dendritic spines tuned to congruent auditory and visual features to exhibit a greater degree of spatial clustering than spines tuned to unrelated pairs, with this enrichment present in reared but not control animals. Such organisation would provide a structural correlate of audiovisual learning, revealing how prolonged exposure to correlated multisensory statistics reshapes synaptic architecture. This clustering is predicted to support dendritic nonlinearities that selectively amplify congruent inputs³⁷, generating multisensory tuning at the somatic level and, in turn, driving the population-level coding shifts and perceptual biases characterised in WP1-WP2. Demonstrating this would show that learned audiovisual associations are physically embedded in, and in principle decodable from, the synaptic micro-architecture of the cortex, providing an unprecedented link between early sensory experience, neural circuit structure, and perceptual function.

Feasibility and environment

The applicant brings over ten years of experience bridging rodent systems neuroscience and computational modelling, with a strong record in investigating visual and multisensory cortical processing through in vivo imaging, electrophysiology, behavioral analysis, and machine learning. The project builds on innovative yet well-established methodologies, including miniaturized two-photon imaging and rodent psychophysics, supported by preliminary results providing clear proof of feasibility. The project will be hosted within the Department of Surgery and Otorhinolaryngology (ORL) service at CHUV, offering state-of-the-art experimental facilities and a shared rodent surgical platform with the auditory neuroprosthetics team of Prof. Lukas Anschuetz. The applicant will also be embedded in *The Sense*, centre, led by Prof. Micah Murray, providing an ideal environment to connect mouse and human multisensory research and to foster cross-species comparisons and translational applications.

Expected results and impact

MUSIC will uncover how unsupervised multisensory experiences forge cross-modal priors, and how these priors are reflected in behavior, cortical population activity, and synaptic architecture, using the mouse model. Although mice and humans differ in developmental timescale and life experience, the computational principles of multisensory learning are likely to be evolutionarily conserved, forming part of the fundamental bedrock of cognition^{18,20}. This makes the mouse an unparalleled experimental system that enables invasive, cutting-edge investigations to uncover mechanisms of sensory learning that remain inaccessible in humans. Moreover, its short lifecycle allows developmental studies that would require many years in other species, while still yielding general insights directly relevant to human perception, cognition, and neurodevelopment. By seizing this opportunity, MUSIC will deliver a breakthrough in our understanding of how the brain builds internal models of the sensory world. In doing so, it will tackle one of cognitive neuroscience's most enduring enigmas: *how do we learn to interpret the world around us?* At the same time, it will spark a genuine interdisciplinary dialogue between neuroscience and artificial intelligence. By evaluating diverse unsupervised learning algorithms and network architectures for their explanatory power over neural and perceptual phenomena, it will enable direct comparisons between machine perception systems and cortical neural networks, an exchange that could drive both AI innovation and conceptual neuroscientific progress. Moreover, by illuminating the physical substrate of implicit sensory memories shaped by early-life experience, MUSIC carries profound significance for understanding the neural basis of learning and memory, extending beyond the field of sensory neuroscience. Unraveling the links between specific changes in synaptic wiring and individuality of perception could add an important piece to the puzzle of how our brains ultimately *make us who we are*.

Beyond basic science, MUSIC bears clear translational implications: by framing neuroprosthetics and sensory substitution as forms of artificial multisensory integration^{31,38}, it will clarify the fundamental learning rules underlying adaptation to this kind of therapeutic intervention. Notably, the proposed image-to-sound mapping is loosely inspired by a real-world visual-to-auditory sensory substitution strategy used in blind patients, where visual information is converted into structured sound patterns that eventually engage visual cortical areas and convey shape information³⁸. In this sense, MUSIC will provide a mechanistic model of a form of cross-modal plasticity similar to that which allows the human brain to reinterpret sensory inputs through alternative channels. This could pave the way for optimised training protocols that accelerate patient adjustment, enhance perceptual outcomes, and *improve quality of life*.

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Curriculum vitae and Track Record

PERSONAL DETAILS

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• Education and key qualifications

- 17/02/2020 - PhD in Cognitive Neuroscience (cum laude)
International School for Advanced Studies (SISSA), Trieste, Italy
Supervisor: Davide Zoccolan
- 12/2015 - M.Sc. in Physics (110/110 summa cum laude)
University of Turin, Torino, Italy
Supervisor: Mario Ferraro
- 07/2013 - B.Sc. in Physics (110/110)
University of Turin, Torino, Italy
Supervisor: Davide Lovisolo

• Current position

- 03/2020 to present - Postdoctoral Researcher, Cognitive Neurobiology Lab
Dept. of Basic Neurosciences, University of Geneva, Switzerland
PI: Sami El-Boustani

• Previous positions

- 09/2016 to 02/2020 - PhD Student, Visual Neuroscience Lab
International School for Advanced Studies (SISSA), Trieste, Italy
PI: Davide Zoccolan
- 12/2015 to 09/2016 - Research Fellow, Visual Neuroscience Lab
International School for Advanced Studies (SISSA), Trieste, Italy
PI: Davide Zoccolan
- 01/2013 to 06/2013 - Research Fellow, Cellular Neurophysiology Lab
Dept. of Life Sciences and Systems Biology, University of Turin, Italy
PI: Davide Lovisolo

RESEARCH ACHIEVEMENTS AND PEER RECOGNITION

Research achievements

I am a systems neuroscientist with a background in physics and a doctoral specialization in visual neuroscience. My work combines behavioral approaches with in vivo recordings (including two-photon calcium imaging and extracellular electrophysiology), optogenetics, and computational modelling to study the cortical basis of multisensory perception and decision-making in rodents. I have extensive experience in large-scale neural and behavioral data analysis and machine learning. I have led multiple interdisciplinary research projects and co-authored nine impactful peer-reviewed publications revolving around four main topics:

Cortical circuits for cross-modal generalization²: I demonstrated how dorsal associative cortical regions in mice co-map visual and tactile spaces, enabling rapid task-rule transfer across sensory modalities. I conducted some of the core experimental work (imaging, behavior, optogenetics), developed

hardware/software for imaging rigs, and built the entire data analysis pipeline for this work. This paper was highlighted in a spotlight article in *Trends in Neurosciences* by Seung-Hee Lee (KAIST) for its broad implications for understanding the neural basis of unified perception of peripersonal space.

Cortical sensory processing across motivational states⁵: I revealed how internal motivational states (such as thirst) modulate sensory processing and decision-making in mouse somatosensory and premotor cortex, providing unprecedented insights into the neural mechanisms of the Yerkes-Dodson law. I led behavioral experiments and carried out extensive data analysis and modeling for this work. These results featured in a *Neuron* spotlight by Jenna A. McHenry (Duke University) for its novel contribution connecting homeostatic drives to goal-directed behavior.

Unsupervised temporal learning in visual development^{8,3}: I provided the first causal evidence that temporal continuity in early postnatal visual experience drives complex cell development in primary visual cortex (V1), corroborating the theory of unsupervised temporal learning. I co-designed the study with PI Davide Zoccolan, performed all in vivo extracellular recordings, and implemented all analyses as the sole non-senior author, demonstrating a strong level of scientific independence during my PhD. On the same topic, I co-authored a review in *Current Opinion in Neurobiology* with Eugenio Piasini and Davide Zoccolan on unsupervised learning of mid-level visual representations, further pointing the road ahead to bridge the fields of unsupervised learning and visual neuroscience.

Shape and motion processing in rat visual cortex^{4,6,9}: I demonstrated nonlinear motion integration and hierarchical shape processing in rat visual cortex as well as the spontaneous perception of global motion by rats, establishing rodents as valid models for key aspects of mid-level vision. I performed most of the in vivo extracellular recordings, behavioral training and testing and implemented all analyses and machine learning modelling for these projects. In both projects, neural representations obtained by in vivo extracellular recordings were compared directly with representations in deep neural networks fed the same visual stimuli as the animals, marking these studies as early and significant contributions to the NeuroAI field.

List of publications in peer-reviewed journals

- 1) Roig-Puiggros S, Guyoton M, Suchkov D, Fortoul A, Fièvre S, **Matteucci G**, Maino E, †Foucher C, Fuciec D, Klingler E, Cappello S, Francis F, Watrin F, Minlebaev M, El-Boustani S, Manent J, Jabaudon D (2025), *Emergence of cortical function in self-organizing heterotopic neurons*, **Nature Neuroscience** (accepted).
- 2) *Guyoton M, ***Matteucci G**, †Foucher C, Getz MP, Gjorgjieva J, El-Boustani S (2024), *Cortical circuits for cross-modal generalization*, **Nature Communications** (DOI:10.1038/s41467-025-59342-9)
- 3) **Matteucci G**, Piasini E, Zoccolan DF (2024), *Unsupervised learning of mid-level visual representations*, **Current Opinion in Neurobiology** (DOI:10.1016/j.conb.2023.102834)
- 4) **Matteucci G**, Bellacosa Marotti R, †Zattera B, Zoccolan DF (2023), *Truly pattern: Nonlinear integration of motion signals is required to account for the responses of pattern cells in rat visual cortex*, **Science Advances** (DOI:10.1126/sciadv.adh4690)
- 5) **Matteucci G**, Guyoton M, Auffret M, Foustoukos G, Mayrhofer J, Petersen C, El-Boustani S (2022), *Cortical sensory processing across motivational states during goal-directed behavior*, **Neuron** (DOI:10.1016/j.neuron.2022.09.032)
- 6) ***Matteucci G**, *†Zattera B, Bellacosa Marotti R, Zoccolan DF (2021), *Rats spontaneously perceive global motion direction of drifting plaids*, **PLoS Computational Biology** (DOI:10.1371/journal.pcbi.1009415)
- 7) ***Matteucci G**, *Riggi M, Zoccolan DF (2020), *A template-matching algorithm for laminar identification of cortical recording sites from evoked response potentials*, **Journal of Neurophysiology** (DOI:10.1152/jn.00033.2020)
- 8) **Matteucci G**, Zoccolan DF (2020), *Unsupervised experience with temporal continuity of the visual environment is causally involved in the development of V1 complex cells*, **Science Advances** (DOI:10.1126/sciadv.aba3742)

- 9) **Matteucci G**, Bellacosa Marotti R, Riggi M, Rosselli FB, Zoccolan DF (2019), *Nonlinear processing of shape information in rat lateral extrastriate cortex*, **Journal of Neuroscience** (DOI:10.1523/JNEUROSCI.1938-18.2018)

* equal contribution, † supervised student

Peer recognition

My contributions have been recognised through several awards and fellowships, most notably the SISSA Best PhD Thesis Award (2021), which honoured the quality and impact of my doctoral research. I was also supported by competitive fellowships and travel grants from the Young Swiss Society for Neuroscience, the Boehringer Ingelheim Fonds, the Italian Ministry of Education and Research, and SISSA. I have been invited to present my work at leading international institutions, including EPFL, the Institut de la Vision in Paris, and the Technical University of Munich, in addition to numerous conference presentations (COSYNE, FENS, SFN, Barrel Meeting, SSN/ySSN). My publications have received editorial highlights, such as spotlights in *Neuron* and *Trends in Neurosciences*, underlining their influence across the field. I am active in the broader research community and as a recognized expert I regularly serve as a reviewer for journals such as *Current Biology*, *eLife*, *eNeuro*, *Scientific Reports*, *Cognitive Computation*, *Computational Brain & Behavior*, and *Frontiers in Psychology*, as well as for COSYNE submissions.

Selected invited seminars and talks

- 1) *Multisensory generalization, motivation and learning in mouse perceptual decision-making*:
Technische Universität München (host: Julijana Gjorgjieva), 07/04/2025, Munich, Germany
- 2) *Cortical circuits for multisensory integration and goal-directed cross-modal transfer learning*:
École Polytechnique Fédérale de Lausanne (host: Carl Petersen), 14/03/2025, Lausanne, Switzerland
- 3) *Cortical circuits for cross-modal generalization*:
1st Fondation Pour l'Audition Horizon Workshop, 10/2024, Mareil-le-Guyon, France
- 4) *Investigation of cortical circuits combining vision and touch*:
Institut de la Vision (host: Olivier Marre), 03/04/2024, Paris, France
- 5) *Modeling the interplay of motivation and learning in mouse perceptual decision-making*:
Swisskors Meeting 2024, 01/2024, Zurich, Switzerland
- 6) *Cortical sensory processing across motivational states during goal-directed behavior*:
SSN-ySSN Meeting, 07/2023, Lugano, Switzerland
- 7) *Sensorimotor representation in mice performing a whisker-based discrimination task*:
Giessbach Meeting, 09/2021, Brienz, Switzerland
- 8) *Sensorimotor interaction in mice performing a whisker-based discrimination task*:
Swisskors Meeting 2021, 01/2021, Geneva, Switzerland

Selected poster communications

- 1) *Cortical circuits for cross-modal generalization*:
1st INCN meeting, 09/2024, Rome, Italy
- 2) *Modeling the interplay of motivation and learning in mouse perceptual decision-making*:
FENS Forum 2024, 07/2024, Vienna, Austria
- 3) *Cortical circuits for goal-directed cross-modal transfer learning*:
SFN Meeting 2023, 11/2023, Washington DC, USA
- 4) *Cortical circuits for goal-directed cross-modal transfer learning*:
FENS Workshop: Structuring Knowledge for Flexible Behavior, 10/2023, Rungsted Kyst, Denmark

- 5) *Cortical sensory processing across motivational states during goal-directed behavior:*
SSN-ySSN Meeting 2023, 07/2023, Lugano, Switzerland
- 6) *Cortical sensory processing across motivational states during goal-directed behavior:*
FENS Forum 2022, 07/2022, Paris, France
- 7) *The causal role of unsupervised temporal learning in the development of V1 complex cells:*
COSYNE 2019, 04/2019, Lisbon, Portugal
- 8) *Dark-reared rats develop higher visual acuity than controls in an orientation discrimination task:*
European Conference on Visual Perception 2018, 08/2018, Trieste, Italy
- 9) *Pattern motion responses in rat visual cortex:*
European Conference on Visual Perception 2018, 08/2018, Trieste, Italy
- 10) *Reverse correlation reveals rich repertoire of spatiotemporal receptive field structures in rat V1:*
Bernstein Conference 2016, 09/2016, Berlin, Germany

ADDITIONAL INFORMATION

Career breaks, diverse career paths and major life events

No career breaks to report. Research productivity during the initial phase of my postdoc in Geneva (2020-2021) was partly affected by COVID-19 restrictions, but normal activity was rapidly resumed.

Other contributions to the research community

I have made sustained contributions to the growth and productivity of my host laboratories and to the broader neuroscience community. In Geneva, I played a foundational role in establishing the El-Boustani laboratory (2020-2025), joining six months after its creation and spearheading the development of software and hardware infrastructure, behavioral and imaging protocols, and acquisition pipelines. This groundwork accelerated the lab's early productivity and enabled key publications in *Neuron* and *Nature Communications*. Earlier, as a PhD student in Davide Zoccolan's group, I made a critical contribution to the ERC project LEARN2SEE by co-developing the Virtual Cage system to manipulate visual experience in rodents, which proved pivotal to uncovering principles of early postnatal visual development, culminating in a publication in *Science Advances*. These experiences have been instrumental in preparing me for the challenge of establishing an independent systems neuroscience laboratory from the ground up, as I propose in this project. I have also invested in mentoring and supervision of six master's students and interns from diverse disciplinary backgrounds, fostering interdisciplinary skills and independence. I have actively contributed to science outreach, including a live interview on Italy's national broadcaster (RAI3), press coverage (*Il Piccolo*), and guided lab visits for students at the University of Geneva. I maintain a personal research website to disseminate accessible content and promote open science. Finally, I have built an international collaborative network spanning experimental and theoretical neuroscience, with active collaborations with Sami El-Boustani (UNIGE), Davide Zoccolan and Eugenio Piasini (SISSA), Julijana Gjorgjieva (TUM), Carl Petersen (EPFL), and Denis Jabaudon (UNIGE). These collaborations have led to joint publications and ongoing projects linking multisensory integration, plasticity, and computational models of learning. These contributions reflect my growing independence as a scientist and my ability to drive interdisciplinary collaborations and mentoring activities, key indicators of my potential for establishing an independent research programme.

Career outlook

Looking ahead, this ERC Starting Grant will provide the decisive step toward establishing my independent research group, dedicated to uncovering the learning principles that shape multisensory perception and experience-dependent plasticity. Building on my combined expertise in systems and computational neuroscience, I aim to develop an internationally visible research programme bridging experimental and theoretical approaches to sensory learning, advancing our fundamental understanding of cortical computation while laying the groundwork for future translational applications in neuroprosthetics and sensory rehabilitation.